

The Price Effects of the Delta/Northwest Merger: Evidence from a Route-Level Panel

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Abstract

In this paper, I use Department of Transportation DB1B Market data to examine the price effects of the 2008 Delta-Northwest merger, over the period of 2006Q1 to 2011Q4. I use a route-quarter panel to analyze difference-in-differences between treated and control routes over time with quarter and route fixed effects. I find that fares on connecting routes rose 2.6% relative to comparison routes, with a confidence interval of 1.8% to 3.3%. Alongside fare changes, passenger volume fell 2.7%, and concentration rose 938 HHI points. This effect appears at full operational integration in 2010, and not at the deal’s announcement or close. While the merger was allowed partly on the premise of minimal overlap, this paper demonstrates that prices still rose on connecting overlap routes.

1 Introduction

How did the Delta and Northwest merger affect prices on shared routes? Fares on connecting routes where Delta and Northwest both operated before the 2008 merger rose 2.6% (95% CI: 1.8-3.3) following full operational integration in 2010, relative to comparable routes. Passenger volume fell 2.7% and market concentration rose 938 HHI points.

Merger retrospectives are critical to determining whether merger review correctly predicts competitive effects. The 2023 Merger Guidelines set structural presumptions regarding concentration, but whether these thresholds are accurate in predicting actual price effects is a question only completed mergers can answer. In the case of the Delta-Northwest merger, the Department of Justice determined that it was “not likely to substantially lessen competition,” due to the “complementary aspects of the airlines’ networks” (U.S. Department of Justice, Antitrust Division, 2008). This paper empirically tests that prediction.

The claim that the networks were complementary is measurable. Of 4,833 nonstop routes, 30 (0.62%) had both Delta and Northwest above a 5% share. This confirms that the nonstop networks were indeed complementary. Consequently, this study focuses on connecting routes, where the networks overlapped on 2,848 routes.

The model utilizes a route-level panel of 18,445 connecting markets over 24 quarters. Route and quarter fixed effects absorb time-invariant route characteristics and common shocks. Treatment is frozen in the pre-period, and treated routes are compared against routes where exactly one of the carriers operated. The effect appears only at operational integration, not at announcement or transaction close. The confidence interval of the estimate is 1.8% to 3.3%, with power to detect effects above 1.08%.

Luo (2014) studies the same merger with a differenced cross-sectional design, and estimates a 2.3% fare increase on connecting overlap markets. This is statistically indistinguishable from the 2.6% I found. Both studies use different identifying assumptions, and arrive at the same conclusion. Additionally, both date the effect to after integration, Luo via a second cross-section, and this paper via a continuous event study. Taken together, the evidence suggests that while the merger was approved in part due to the claim that networks were complementary, fares still rose where the two carriers overlapped.

2 Background

On April 14, 2008, a merger between Delta and Northwest was announced, creating what was the largest airline in the world (Delta Air Lines, 2008). To achieve approval of this merger, the airlines argued that their networks consisted of minimal overlap. In turn, they claimed that this would produce cost savings while not decreasing competition. After a six-month investigation, the U.S. Department of Justice concluded that the merger was “not likely to substantially lessen competition” and that consumers would benefit from combining “the complementary aspects of the airlines’ networks” (U.S. Department of Justice, Antitrust Division, 2008).

The merger took effect in stages over the span of two years. Following the initial announcement, shareholders approved the merger on September 26, 2008, leaving the final decision to the U.S. Department of Justice, which closed its investigation a month later on October 29, allowing the transaction to close (U.S. Department of Justice, Antitrust Division, 2008). From there, Delta and Northwest began work on integrating their operations and systems. By December 31, 2009, the FAA issued a single operating certificate for the combined airline, and by January 31, 2010, they completed the merger by fully integrating the reservations system, retiring the Northwest brand (Luo, 2014).

Between the transaction’s close in October 2008 and full reservations system integration in January 2010, the airlines were legally a single firm, but operationally two. From these dates, I establish three distinct period regimes. First, the pre-period is defined as any date through 2008Q1. This was the last quarter before the merger was announced, and is chosen because the announcement itself could have triggered repricing. Second, the transition period is defined as 2008Q2 to 2009Q4. This spans the merger’s announcement to the full operational integration of both airlines. Finally, the post-period is defined as 2010Q1 onward, the first quarter in which the reservation systems finally merged. These boundaries were determined entirely from the timeline

of the merger and fixed before any estimation.

3 Data

The data is sourced from the Department of Transportation DB1B Market data, a 10% sample of domestic flight itineraries reported quarterly. The selected sample period is from 2006Q1 to 2011Q4. Reported fares are prorated across directional markets by distance, so Market Fare is a one-way equivalent fare. Additionally, a DB1B connecting itinerary generates a single market record with two boardings, so market records are not enplanements.

A market is defined as a non-directional airport pair, meaning a route like ATL-DTW is treated the same as DTW-ATL. This is due to the competitive situation being identical for both combinations, with the same competitors and market power. Moreover, because standard errors are clustered by route, including both as separate routes would understate uncertainty. This suppression of directionality is also utilized by Luo (2014).

Market shares are built using the ticketing carrier rather than the operating carrier. Competition concerns who sets the prices and sells the ticket. Often Delta Connection flights will be operated by regional affiliates under separate codes but ticketed as DL. Using the operating carrier would fragment Delta's presence across different regional affiliate codes and leave it understated on routes of interest.

The following filters are applied to the collected data, with the percentage of records removed from each recorded in parentheses.

1. Records missing fare, passenger, carrier, or route information (0.00%).
2. Records indicated to have been sold with bulk fares, where recorded fare from consolidator and tour-operator tickets is not the price any passenger paid (0.07%).
3. Records indicated to have a mid-ticket ticketing-carrier change, since the selling carrier is not well defined for the itinerary as a whole (4.71%).
4. International itineraries and any routes not fully operated in the contiguous United States, due to different cost structures as well as not being in the network the merger concerned (5.83%).
5. Itineraries with more than five coupons in a directional market, because high-coupon itineraries have the most noise in their prorated fares (0.003%).
6. Fares outside the range of \$25-\$2500 (Floor: 5.40%, Ceiling: 0.015%). Applied last to limit implausible fares unexplained by previous filters. The floor also helps to prevent the logarithmic transformation from severely stretching the left tail.

In total these filters dropped approximately 15.2% of all records.

After being collapsed to a route-carrier level panel, all route-quarters with fewer than 20 sampled passengers are dropped. This results in 53% of markets being removed, but only 1.1% of passengers. While the 53% figure looks high alone, the 1.1% helps to justify the threshold, as it is removing thin markets while maintaining nearly all measured travel.

I chose to analyze markets separately by itinerary type. This is due to the fact that nonstop and connecting service are distinct products, with different sets of competitors between them. I decided to make connecting markets the primary specification in this study. Of the 4,833 nonstop markets remaining after filtering, only 30 had joint Delta and Northwest presence each above the 5% share threshold. This quantifies the complementary-network rationale used to pass the merger, and is too few treated units to support cluster-robust inference.

Before the treatment definition, the filtered panel consisted of 18,445 connecting and 4,833 nonstop markets across 24 quarters. Numbers following treatment classification are observable in Table 1. It can be noticed that treated routes are less concentrated, with a lower HHI and higher carrier count. This can be attributed to the requirement of having two carriers both over a 5% share in the market, which naturally guarantees a more competitive market. It is also noticeable that treated routes and Control A routes are close, with similar fares and distances. Control B, on the other hand, differs on many dimensions, including fewer carriers, higher HHI, lower fares, and less traffic.

Table 1: Summary statistics, pre-merger period (connecting markets)

group	control A	control B	treated
Number of Routes	5328	2521	2848
Fare Mean	220.01	208.19	219.76
Pax per Quarter Mean	179.16	134.83	227.68
Pax per Quarter Std.	283.07	209.34	313.38
Number of Carriers Mean	3.44	2.54	4.53
Number of Carriers Std.	1.80	1.56	1.64
HHI Mean	5686.64	7295.50	4038.08
HHI Std.	2096.61	2155.69	1262.06
Distance Mean	1313.96	1174.37	1263.96
Distance Std.	603.55	590.91	550.20

Notes: Pre-merger period is 2006Q1–2008Q1. **Fare Mean** is passenger-weighted; all other means are unweighted across routes. Standard deviations are across routes within group.

4 Empirical Strategy

To estimate the merger’s effects on fares, I estimate the following specification

$$\ln(\text{fare}_{rt}) = \beta (\text{Treated}_r \times \text{Post}_t) + \alpha_r + \gamma_t + \varepsilon_{rt} \quad (1)$$

where r indexes route, t indexes quarters, α_r is route fixed effects, and γ_t is quarter fixed effects. Here, β is the coefficient of interest.

The identification comes entirely from two sets of fixed effects. Route fixed effects make each route its own control, which allows for all time invariant characteristics like distance, slot constraints, endpoint city size, business-vs-leisure mix, to be absorbed. On the other hand, quarter fixed effects allow for everything common across routes in a period to be absorbed: fuel prices, the recession, seasonality. What remains is the differential movement of the overlap routes in the merger. The resulting consequence of this method is that the specification is based on almost no control variables, since the fixed effects completely cover all confounders that are constant within route or common across routes. This does not come without boundaries, though, as anything that varies within route, over time, and differently between treated and control routes is not absorbed, which is a narrow class of confounders, but not an empty one; this is discussed further in Section 7.

This design requires that, if the merger were to not have happened, fares on the treated and control routes would have followed parallel paths. While parallel path trends are unobservable after the merger, the observable implication is parallel movement before it. These pre-trends can be observed in Figure 1, with Control A and the treated group following each other tightly before the merger, and Control B trending above both.

A route is classified as treated if Delta and Northwest each accounted for at least 5% of its pre-period passengers, and met a minimum of at least six of nine observed pre-period quarters. The 5% metric was determined using a pooled share of passengers calculated for all carriers on each of their given routes over 2006Q1–2008Q1. To avoid conditioning on an outcome of the merger, specifically Delta’s subsequent ability to rationalize its network and exit routes, treatment status is frozen using pre-period data.

Two comparison groups are constructed, Control A and Control B. Control A was built out of routes where exactly one of the carriers exceeded the 5% threshold; Control B is where neither did. These two control groups allow me to estimate what might have happened in the period without the merger. With Control A, the merger didn’t change competition, so it differences out shocks specific to the two carriers. Control B on the other hand does not, and includes structurally different markets.

On the same data and treated routes, the two control groups yield opposite estimates. Against Control A, fares rose approximately 2.57%, while against Control B, they fell 1.70%. I chose Control A as the primary control group solely due to the pre-period evidence, and this choice was made before estimating either specification. Pre-period gaps against Control A hover near zero, with a mean of -0.003, and max absolute deviation of 0.016, as quantified in Figure 2. On the other hand, gaps against Control B drift from -0.01 to -0.05 pre-merger, and widen to about -0.06 by 2011Q4. With parallel pre-trends being the identifying assumption, Control A clearly satisfies this criterion over Control B. Table 1 further corroborates this choice, as Control B routes differ from treated routes in carrier count, concentration, and traffic, in ways that Control A does not. While Control

A is the primary control group chosen, Control B is reported throughout.

To ensure a clean estimation of the merger’s effect, three key restrictions are applied to the final sample. First, all transition quarters (2008Q2–2009Q4) were dropped, where, as explained in Section 2, the carriers were legally one firm but operationally two. Second, routes present in fewer than six pre-quarters are excluded, as without enough pre-period presence, they cannot be classified. Finally, routes where UA and CO, or WN and AirTran, both held $\geq 5\%$ pre-merger share are excluded from all groups. Both of these mergers closed inside the post-period, and including them would fail to construct a comparison route experiencing no merger, and fail to isolate Delta–Northwest merger fare changes in treated routes. Dropping these alternative merger routes removes 655 treated (18.7%), 640 Control A (10.7%), and 97 Control B (3.7%) connecting routes.

Finally, the analysis focuses on three outcome variables: log fare, log passengers, and market concentration (HHI). HHI is the first stage, as it verifies if the merger actually changed the market structure. Passenger volume is included to help distinguish fare changes that may have been driven by reduced capacity. Because fares on the same route are highly correlated over time, in all estimations I cluster standard errors at the route level to avoid overstating precision (Bertrand et al., 2004).

5 Results

Figure 1 plots mean log fare by group across the sample period, normalized to 2008Q1, and restricted to routes present in all twenty-four quarters. Routes are balanced here to prevent the series from mixing fare changes with route entry and exit. All three groups follow the same general pattern, tracking the 2009 recession trough together. Control A and the treated group move tightly together before 2010, while Control B trends above both, especially through 2011.

Table 2 reports the baseline estimates, including fare, passenger volume, and HHI estimated effects. Following full operational integration, relative to Control A, fares on treated connecting routes rose 2.57% (95% CI: 1.80 to 3.35), passenger volume fell by 2.67% (95% CI: -4.41 to -0.91), and HHI increased by 937.93 points (95% CI: 878.69 to 997.17).

The three outcomes move in a pattern characteristic of a supply-side change rather than a shift in demand. Prices rose while quantities fell, whereas a demand shock would move both in the same direction. Market concentration (HHI) on treated markets rose from 4,038 to roughly 4,976, a 23.2% increase. According to the Merger Guidelines issued by the U.S. Department of Justice and Federal Trade Commission (2023), markets above 1,800 HHI are considered highly concentrated, with increases of more than 100 points being significant. The observed increase is nearly 9.4 times that threshold, showing that the treated markets became considerably more concentrated following the merger.

The nonstop estimates are reported in Table 2 for completeness, but cannot support inference. Nonstop routes against Control A see an 18.56% increase in fares (95% CI: 5.95 to 32.66). Because this estimate rests on only 30 treated routes, the cluster-robust standard errors are unreliable. Luo

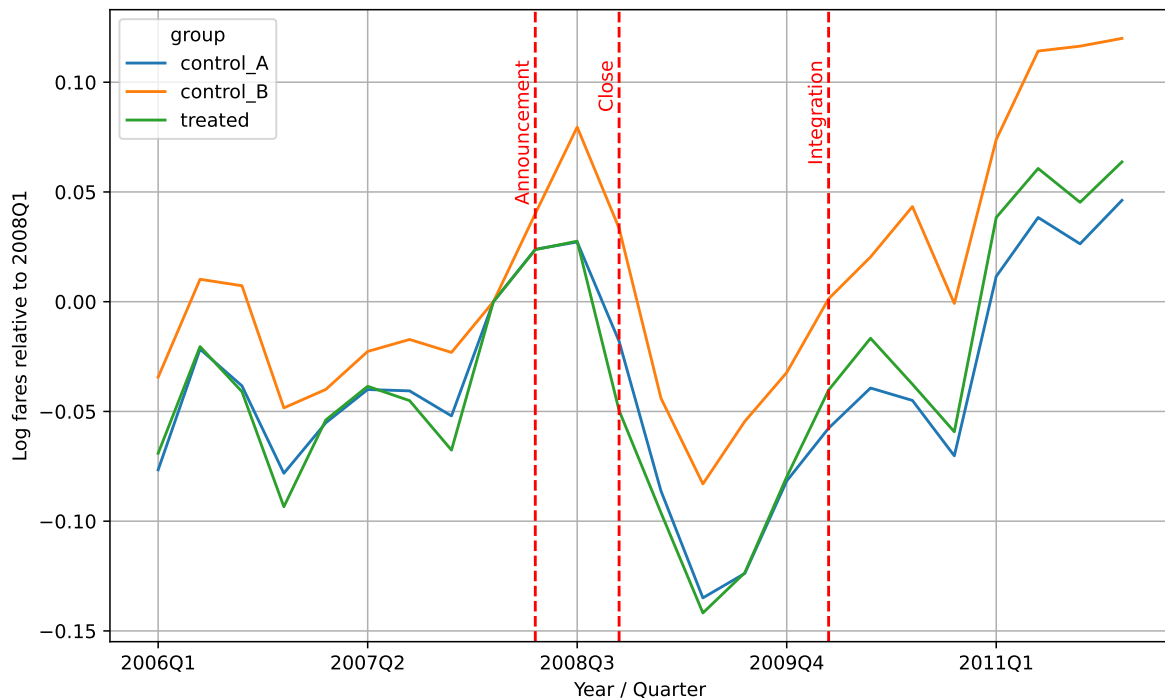


Figure 1: Mean log fare by group, normalized to 2008Q1 (connecting markets, balanced panel)

(2014) reaches the same conclusion from a sample of eight overlap routes.

Table 2: Difference-in-differences estimates

Specification	Fare (%)	Passengers (%)	HHI (points)
connecting / Control A	2.57	-2.67	937.93
connecting / Control B	-1.70	-1.19	1163.50
nonstop / Control A	18.56	12.99	1391.59
nonstop / Control B	9.02	32.95	1635.76

Notes: Route and quarter fixed effects in all columns. Transition quarters (2008Q2–2009Q4) excluded.

Figure 2 reports quarter-by-quarter estimates relative to 2008Q1. Differences remain relatively close to zero throughout the announcement as well as the pre-period and transition period. Fares begin to noticeably rise in 2010Q1, and average +0.022 across the eight post-integration quarters. This demonstrates that fares responded to full operational integration rather than the deal itself. The 2008Q4 coefficient does exhibit a sharp drop, but falls inside the transition window, which is excluded from the baseline specification, and runs opposite to the estimated effect.

Pre-period coefficients average -0.003 with a mean 95% confidence interval half-width of 0.008, so deviations from parallel trends larger than around 0.8% would have been detectable. The estimated effect is 2.6%, so the detection floor is roughly a third of what is observed. Three of the eight pre-period coefficients are individually significant, but they do not trend consistently. Roth (2022) shows that conventional pre-trends tests are often underpowered against violations large

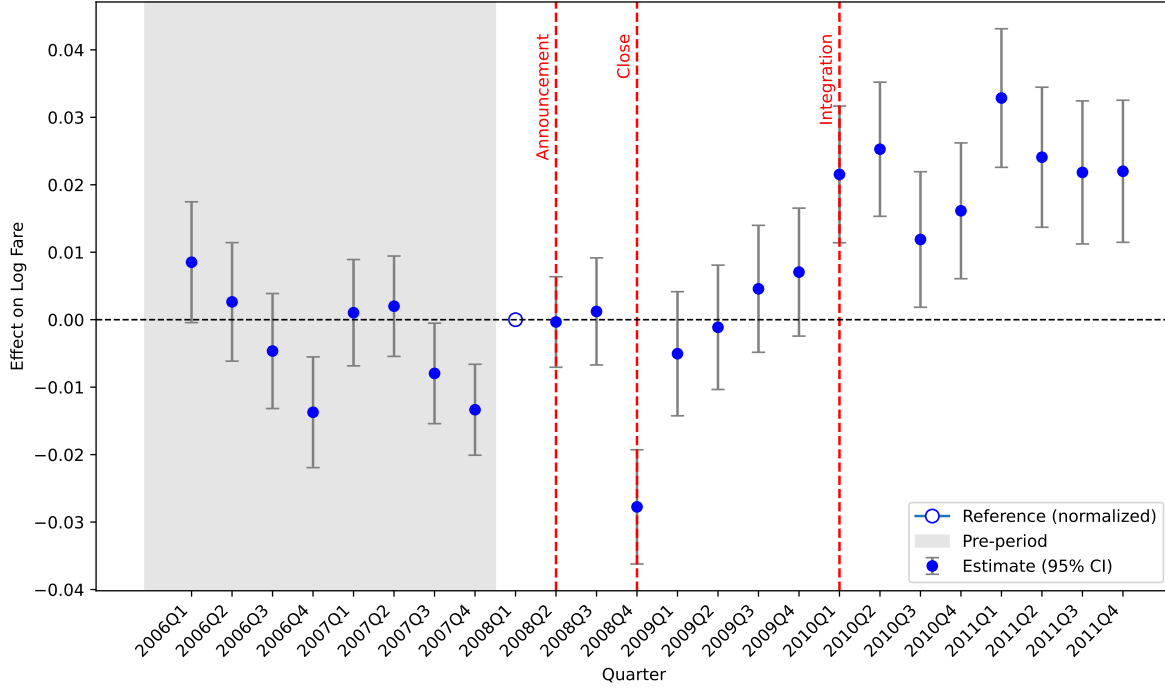


Figure 2: Event-study estimates, connecting markets vs. Control A

enough to generate bias of similar size to the estimated effect. Because differential trends larger than 0.8% per quarter would be detectable, the confidence bands are narrow enough to rule out violations that would meaningfully bias the estimated 2.6% effect.

Table 3 reports robustness test specifications and results. The estimates range from 1.97% to 2.57% across specifications, with all confidence intervals overlapping. The placebo merger yields an estimated effect of -0.32% (95% CI: -0.82 to 0.18), with power to detect changes of more than 0.72%. This suggests the design is not capturing spurious trends, with the placebo interval excluding the baseline estimate entirely. Weighting the original specification by passenger volume results in an estimated effect of 1.97% (95% CI: 1.09 to 2.87). This indicates that fare increases were larger in smaller markets, so the average market saw more of an increase than the average passenger.

Table 3: Robustness

variant	n	coef (%)	p	CI (%)
Baseline	122997	2.57	<0.001	[1.80, 3.35]
Confounded Allowed	143575	2.30	<0.001	[1.61, 3.00]
Balanced Routes	97220	2.27	<0.001	[1.44, 3.12]
Placebo Merger (2007)	63371	-0.32	0.214	[-0.82, 0.18]
Passenger Weighted	122997	1.97	<0.001	[1.09, 2.87]

Notes: Each row re-estimates the baseline specification changing one choice.

The confidence interval rules out fare increases below 1.8% and above 3.3%. Additionally,

because the baseline also has 80% power to detect effects above roughly 1.08%, the 2.57% result demonstrates a precisely estimated impact.

6 Comparison to Luo (2014)

Luo (2014) examines the same merger using the same underlying data source, with a different empirical design. When analyzing connecting routes where Delta and Northwest both competed pre-merger, Luo determines that fares increased by approximately 2.33% (coefficient 0.023, standard error 0.006). I find an increase of 2.57% (coefficient 0.025, standard error 0.004). The difference between the two estimates is around 0.002 log points with a standard error of 0.007. This is not a statistically significant difference.

In Luo’s study, he estimates a differenced cross-sectional regression between two quarters, identifying the merger effect using a dummy variable for pre-merger presence, while directly controlling for changes in the number of low-cost and legacy competitors, adjacent and potential LCC competition, and local unemployment. I estimate a panel regression over twenty-four quarters, identifying the effect from route and quarter fixed effects, with no direct competition controls. Despite the two designs having such different identifying assumptions, both arrive at nearly indistinguishable results, strengthening the validity of both findings.

Aside from estimating the effect itself, both studies also agree on when exactly the effect emerged. Luo re-estimates the results for his connecting model using 2007Q3 as the pre-merger period and 2009Q3 as an intermediate period, which lands where the merger had closed, but operational systems like the reservation network had not yet been integrated. Here, Luo finds an overlap coefficient of -0.003 with a standard error of 0.005, statistically indistinguishable from zero. Luo (2014, p. 39) concludes from this that market power effects started to dominate “only after the merger was finalized.” This matches what I find, represented in Figure 2 continuously instead of just at two points. Differences between the treated group and Control A stay near zero through 2009Q4, with the 2009Q3 coefficient being +0.005, and step up starting in 2010Q1, averaging around +0.022 in the post period. Using the panel design, I am able to trace the onset to the quarter the reservation systems merged.

A few more scope differences exist between Luo’s and my design. First are alternative sample restrictions. Luo requires at least 900 passengers per quarter for a market to be included, whereas I require 20 sampled passengers per route-quarter. From this, Luo analyzes 3,364 airport-pairs, and I analyze 18,445. Second, in his treatment definition, Luo’s overlap dummy requires Delta to continue serving the route following the merger. In my paper, treatment is frozen in the pre-period to avoid conditioning on the outcome of the merger. Finally, Luo identifies 8 nonstop routes, and I find 30 under a 5% share threshold. Both papers reach the same conclusion of nonstop markets not being able to support inference, with Luo getting a coefficient of 0.011 with a standard error of 0.058, while I find a coefficient of 0.170 with a standard error of 0.057.

A notable advantage Luo’s paper has over this one is his use of explicit controls for changes in

competition and unemployment. Because of this difference, Luo is able to break the fare change down into its components, showing that while the merger contributed +2.3%, entry by other legacy and other low-cost carriers contributed roughly -1.1% and rising unemployment contributed -3.0%. As a result, the actual fare change on connecting routes was near zero. Both estimates reported are *ceteris paribus* effects and not the actual price change consumers would have felt. The panel design cannot decompose the change like Luo’s, but it can test the identifying assumption directly, date the onset to a single quarter, and bound the effect between 1.8% and 3.3%.

7 Limitations

The most serious risk to identification is a confounder that the fixed effects cannot absorb. Many of the treated routes are concentrated on Detroit and Minneapolis hubs, and Detroit in particular experienced a severe localized shock as the automotive industry collapsed during the transition window. This in turn caused local demand to fall on many treated routes for reasons unrelated to the merger. This demand shock varies within route, across time, and differently between treated and control routes, meaning it can survive both fixed effects. In his study, Luo (2014) includes a change-in-unemployment control built from Metropolitan Statistical Area data from the Bureau of Labor Statistics. Luo weighted these local unemployment rates by route endpoint populations, yielding an estimated coefficient of -0.007. Based on this, his decomposition attributes roughly -3.0% of the overall connecting fare change to rising unemployment. It is worth noting that a local demand shock on treated routes would bias the estimate downward, so if the auto industry collapse does have an effect, it makes the 2.6% estimate conservative. Regardless, adding MSA-level unemployment as a time-varying covariate would be a valuable extension for future research.

Some limitations also arise from the measurements in the available DB1B market data. First, the data from which route-quarter means are estimated is a random 10% sample. This introduces high variance and outlier sensitivity in thin markets, widening standard errors. Second, round-trip itineraries are prorated across the two directional markets. This mixes true one-way fares with half-round-trip fares. Finally, any extra fees customers might have paid on top of their ticket purchase are not accounted for. This includes things like checked-bag and change fees, which started becoming standardized in the industry during 2008. Quarter fixed effects can only capture this if airlines unbundled on similar schedules. Otherwise, if Delta and Northwest started incorporating these changes at different times compared to other carriers, a portion of the estimated effect could reflect a change in what the fare actually includes rather than a change in base prices.

Possible limitations also arise from definitional choices I made when creating the cleaned dataset. I defined markets strictly as airport pairs, not city pairs. Airport pairs ignore competition between airports in the same city, overstating concentration. Another group of possible limitations arises from my choices of thresholds. This includes a coupon cap of 5, a fare-trim ranging from \$25-\$2500, and a 5% Delta and Northwest overlap threshold for treated routes. Each of these choices has possible effects on the outcome that are not accounted for in the analysis.

Two further limitations affect what the estimates actually capture. At only 30 treated routes, the nonstop estimates cannot support inference. The study is also only based on a single merger in one period and industry. To truly validate the claim that complementary-network mergers produce small positive price effects, more retrospectives are required.

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